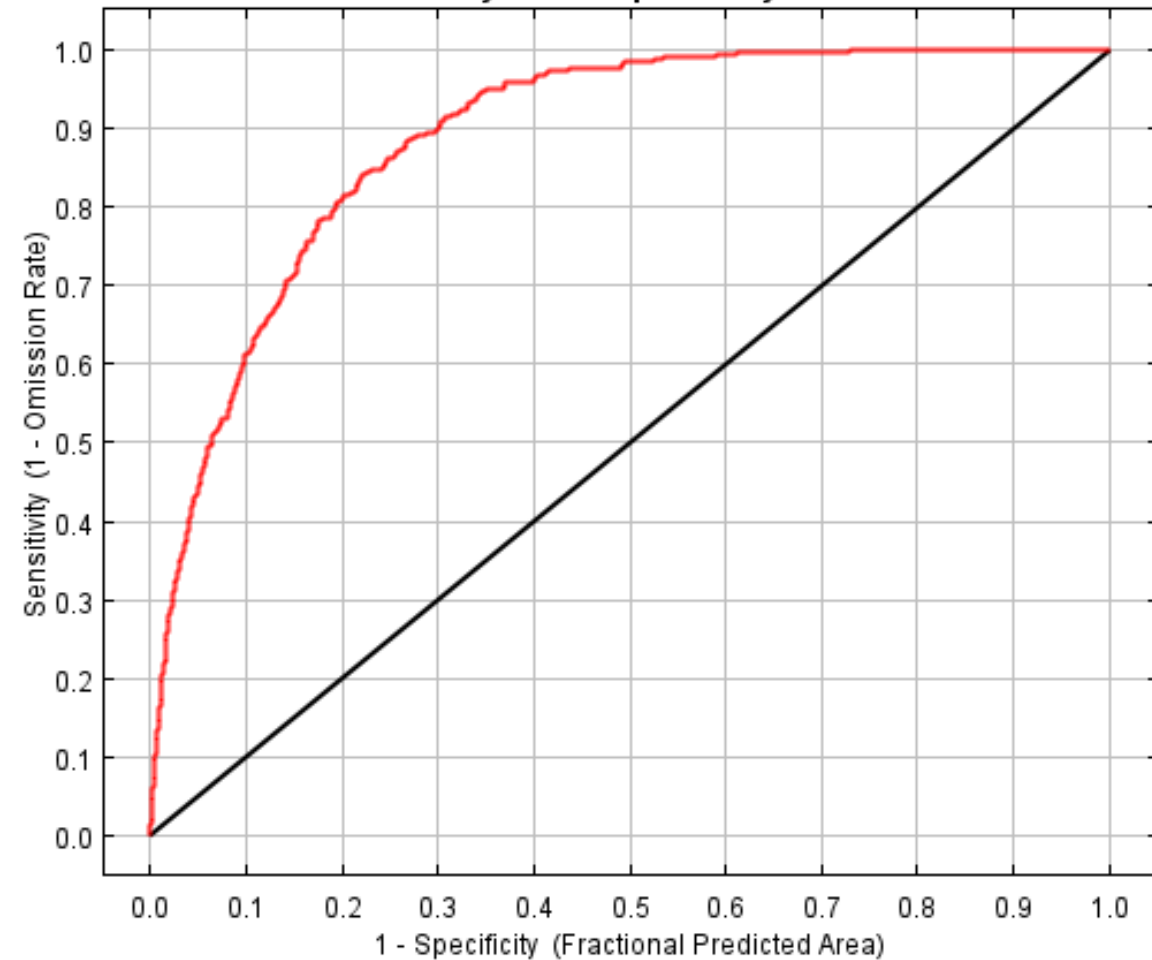


Model Threshold and Evaluation

Gengping Zhu

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Sensitivity vs. 1 - Specificity for YSSB



Training data (AUC = 0.888) ■
Random Prediction (AUC = 0.5) ■

AUC value

Area Under the ROC Curve

ROC curve

Receiver Operating Characteristics

Input data for model evaluation

- Testing point (long, lat, CSV)

data splitting strategy

- **Model suitability/binary predictions:**

Threshold dependent approach

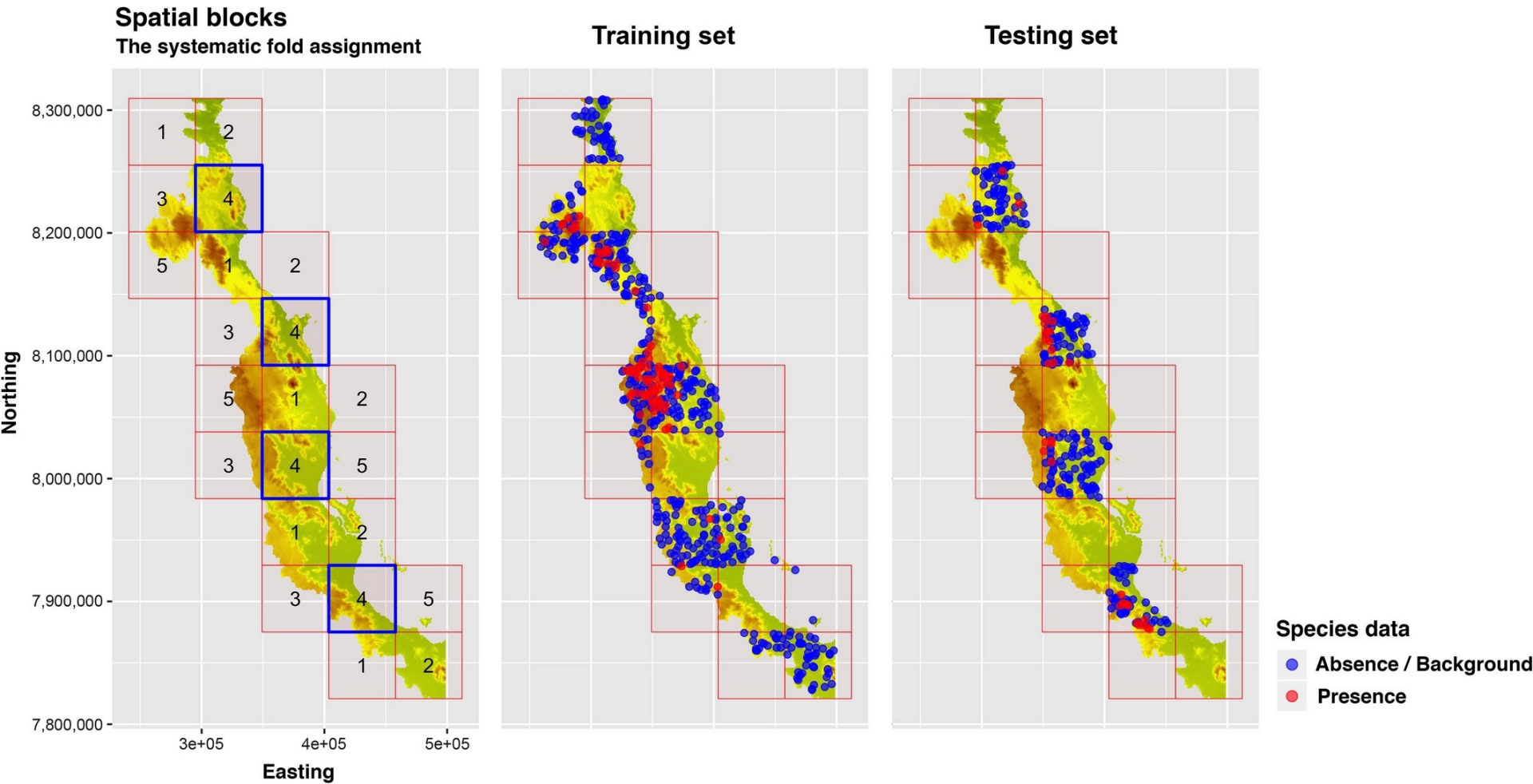
-> binary prediction

Threshold independent approach

-> suitability prediction

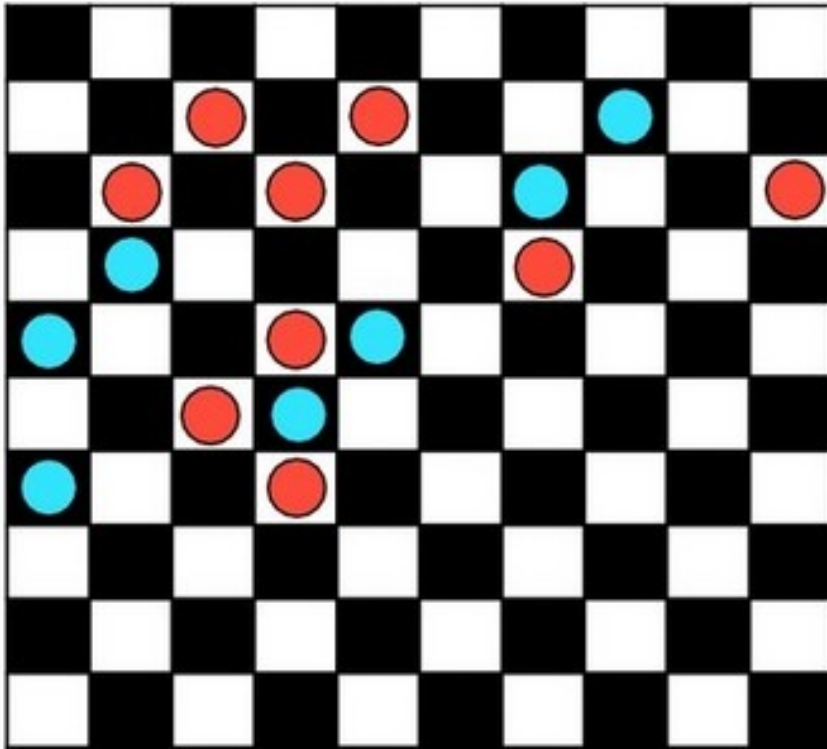
-> they still need “binary” predictions but summarize the metrics across the spectrum of thresholds.

Method 1: Spatial block cross validations

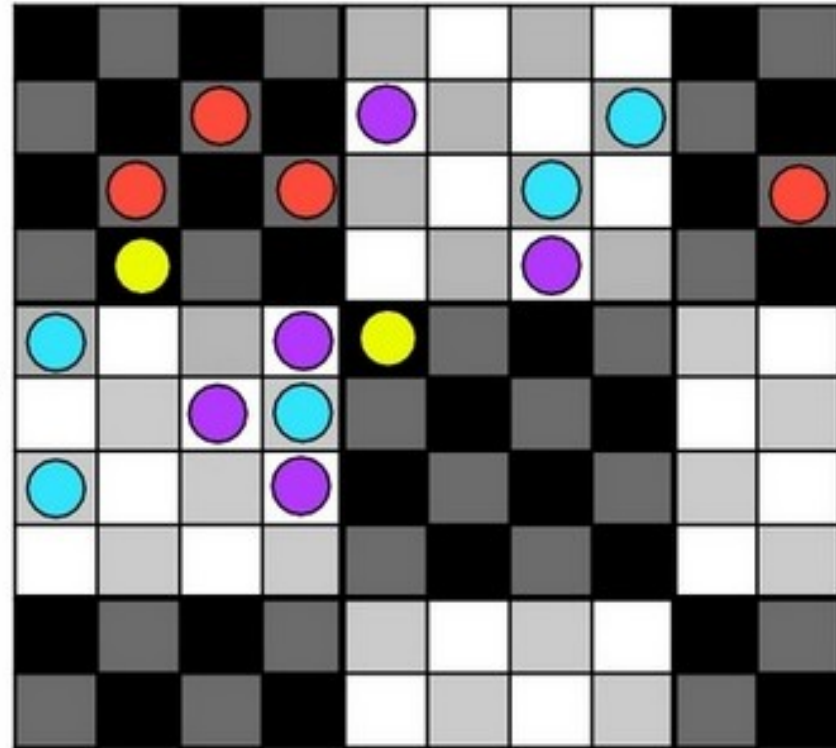


Method 2: Spatial checkerboard

(c) 'Checkerboard1'



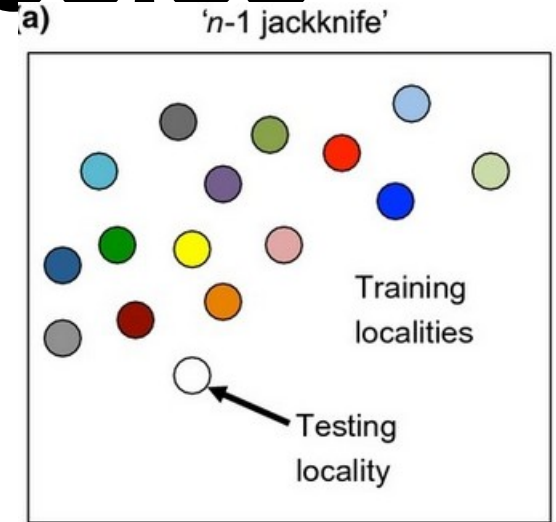
(d) 'Checkerboard2'



R package: ENMeval

Rare, endemic species

- Leaving-one-out method;
Proposed by Richard Pearson et al. 2007
lowest presence threshold (LPT)



leave-one-out (jackknife)

-
- **Ensemble of small model;**
Proposed by Guisan group
ecospat package

Model evaluation overview

Threshold dependent approach

- Omission error (**Type I**), Commission error (**Type II**);
- Sensitivity, Specificity, Kappa, True Skill Statistic (TSS)
- ...

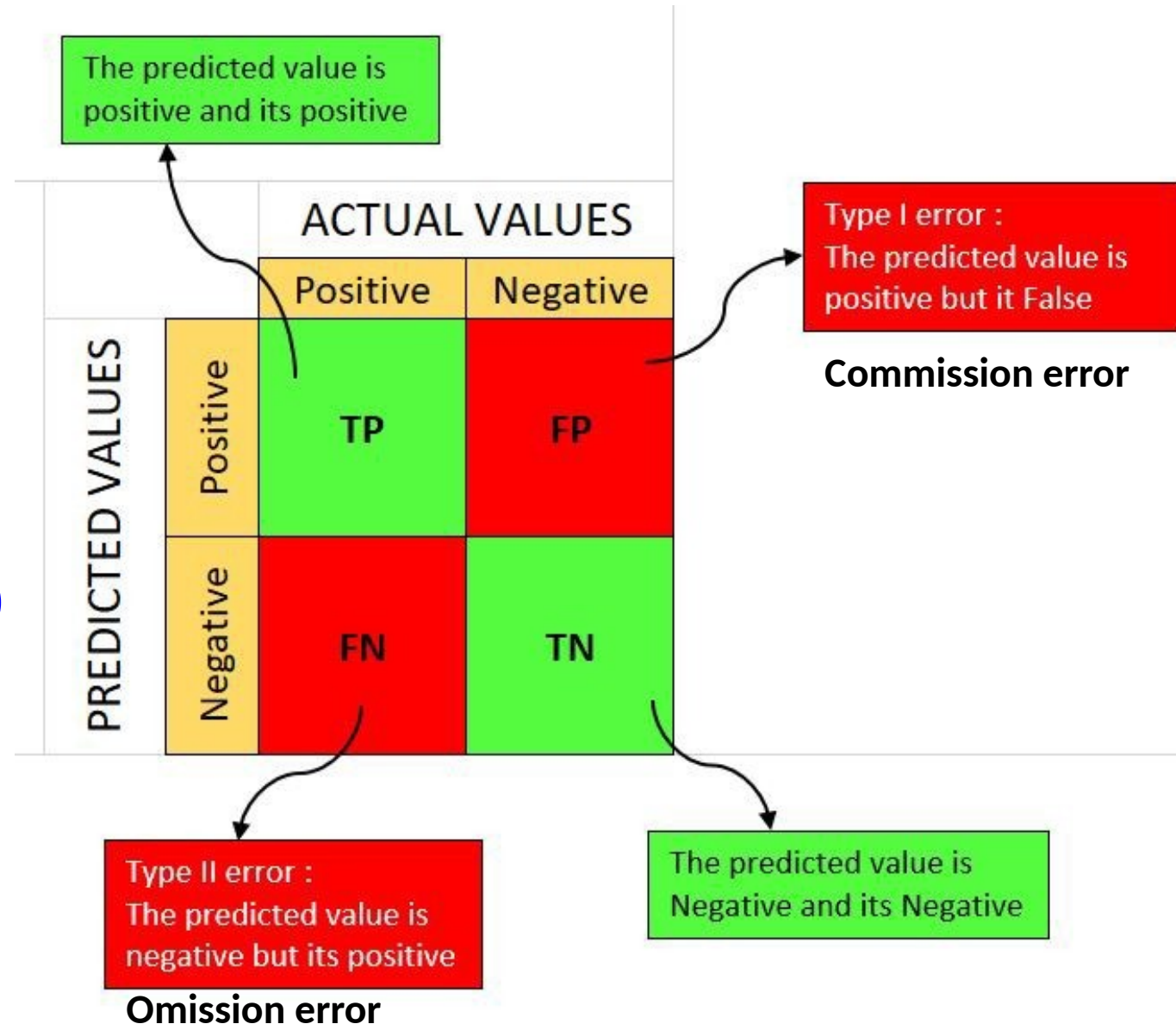
Threshold independent approach

- AUC, AUC ratio, Boyce index

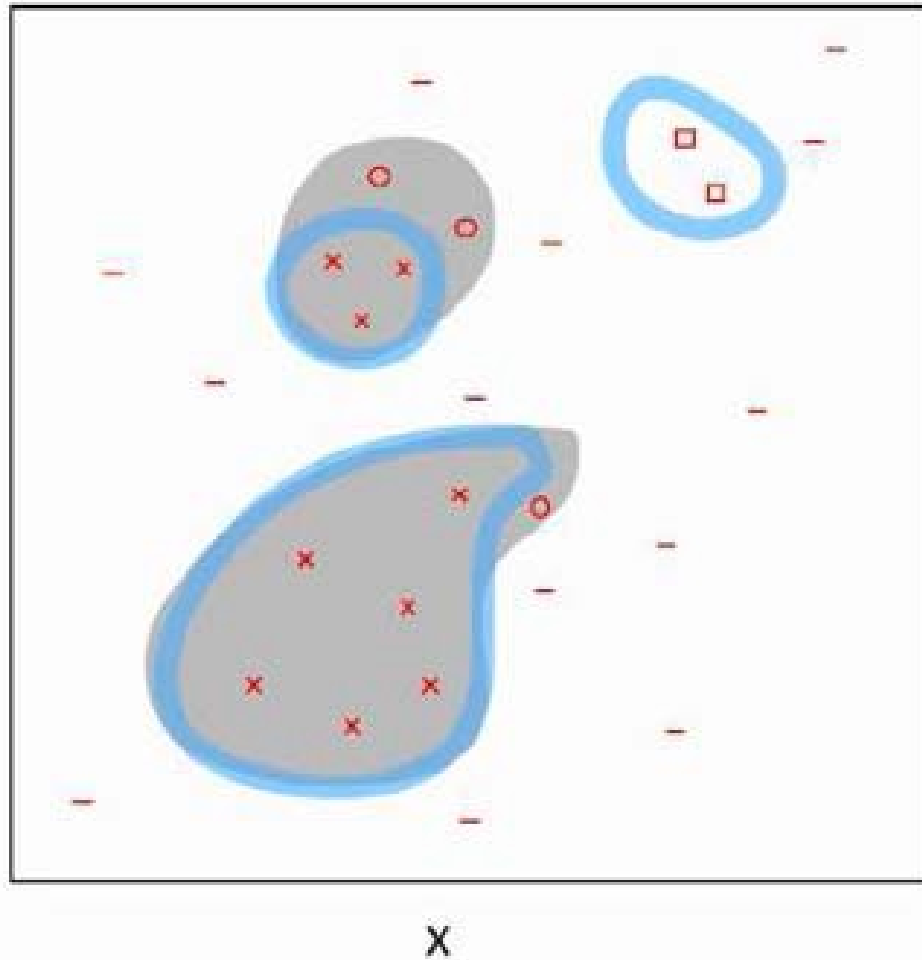
Present-only model:

- AUC ratio, Boyce index

Confusion Matrix for Binary Data



Geographical space



-  Actual distribution
-  Species distribution model
-  True positive
-  False positive
-  False negative
-  True negative

Sensitivity: true positive rate (TPR):

$$\text{Sensitivity} = \text{TPR} = \frac{TP}{TP + FN}$$

Specificity: true negative rate (TNR):

$$\text{Specificity} = \text{TNR} = \frac{TN}{TN + FP}$$

True Skill Statistic (TSS)

$\text{Sensitivity} + \text{Specificity} - 1$
ranges: -1 to 1

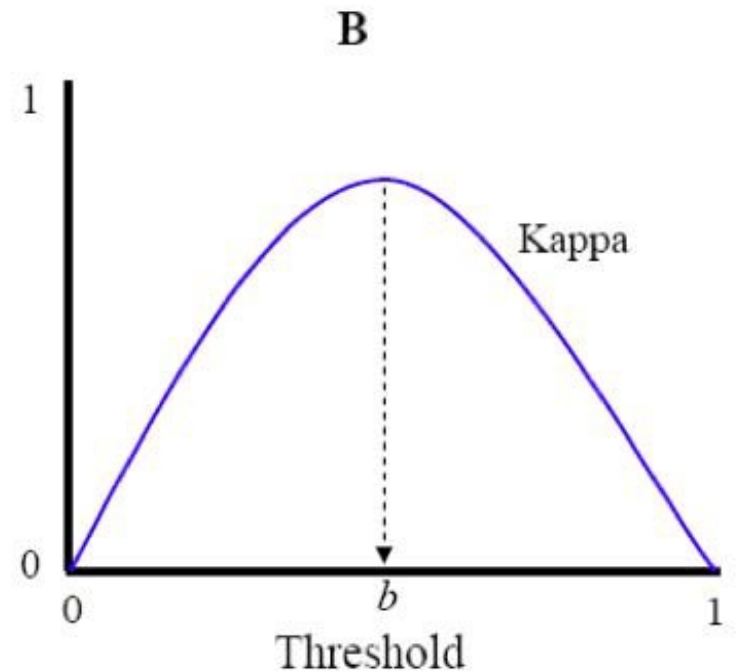
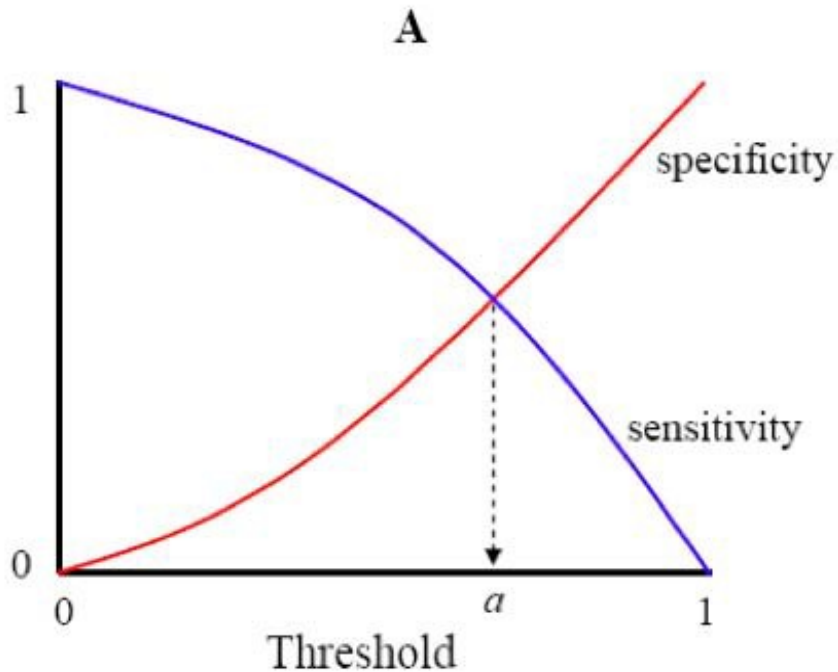
Widely used in medical diagnostic tests

Confusion Matrix

	Actually Positive (1)	Actually Negative (0)
Predicted Positive (1)	True Positives (TPs)	False Positives (FPs)
Predicted Negative (0)	False Negatives (FNs)	True Negatives (TNs)

Kappa

- measure corrects the overall accuracy of model predictions by the accuracy expected to occur by chance, ranging from -1 to $+1$.



Threshold

- value to use as the “cut-off”
- apply a given cut-off value to model and output a raster with binary values for presence and absence.

Cumulative threshold	Cloglog threshold	Description	Fractional predicted area	Training omission rate
1.000	0.038	Fixed cumulative value 1	0.593	0.007
5.000	0.152	Fixed cumulative value 5	0.426	0.026
10.000	0.254	Fixed cumulative value 10	0.342	0.060
0.175	0.011	Minimum training presence	0.728	0.000
13.940	0.329	10 percentile training presence	0.300	0.099
28.793	0.493	Equal training sensitivity and specificity	0.195	0.195
23.938	0.447	Maximum training sensitivity plus specificity	0.224	0.156
1.800	0.064	Balance training omission, predicted area and threshold value	0.534	0.010
7.815	0.211	Equate entropy of thresholded and original distributions	0.373	0.043

Threshold---testing records

Description
Fixed cumulative value 1
Fixed cumulative value 5
Fixed cumulative value 10
Minimum training presence
10 percentile training presence
Equal training sensitivity and specificity
Maximum training sensitivity plus specificity
Balance training omission, predicted area and threshold value
Equate entropy of thresholded and original distributions

Fixed value

Present-only

Present-background
/pseudo-absent

The “Balance” threshold minimizes:

$6 * \text{training omission rate} + .04 * \text{cumulative threshold} + 1.6 * \text{fractional predicted area}.$

Threshold-provided by **PresenceAbsence** package

```
> optimal.thresholds(comb)
      Method      pred
1   Default 0.50000000
2   Sens=Spec 0.45000000
3 MaxSens+Spec 0.43000000
4   MaxKappa 0.73000000
5   MaxPCC 0.99000000
6 PredPrev=Obs 0.89000000
7   ObsPrev 0.03334412
8   MeanProb 0.22599561
9   MinROCDist 0.43000000
10  ReqSens 0.38000000
11  ReqSpec 0.52000000
12   Cost 0.99000000
```

Minimum training presence (MTP)

- ◆ This threshold finds the lowest predicted suitability value for an occurrence point.
- ◆ it assumes that the least suitable habitat at which the species is known to occur is the minimum suitability value for the species.
- ◆ all testing points would fall within the area of the binary prediction.
- ◆ Suitable for rare and endemic species that have limited occurrence records.

10th percentile training presence

- ◆ The P10, on the other hand, is a threshold which omits all regions with habitat suitability lower than the suitability values for the lowest 10% of occurrence records.
- ◆ It assumes that the 10% of occurrence records in the least suitable habitat aren't occurring in regions that are representative of the species overall habitat, and thus should be omitted.
- ◆ This threshold omits a greater region, and is more conservative than the MTP.

What is the advantage of MTP and 10th training thresholds?

-> they have no requirement of absence testing data

How to calculate MTP, 5th, and 10th training threshold?

For example, we have 100 testing points:

Increasing

1	long	lat	Maxent Cloglog value
2	-74.75376669	39.65318847	0.000000741
3	-74.75168335	39.65318847	0.000000864
4	-74.74751668	39.65318847	0.000001209
5	-74.75376669	39.65110513	0.000001423
6	-74.75168335	39.65110513	0.000002122
7	-74.74960002	39.65110513	0.000002966
8	-74.74751668	39.65110513	0.000005955
9	-74.74543334	39.65110513	0.000009865
10	-74.75585003	39.6490218	0.000015039
11	-74.75168335	39.6490218	0.000024111
12	-74.74751668	39.6490218	0.00004452
13	-74.74543334	39.6490218	0.000073087
14	-74.75168335	39.64693846	0.00014156
15	-74.74960002	39.64693846	0.000340998
16	-74.74751668	39.64693846	0.00070711
17	-74.74543334	39.64693846	0.00128156
18	-74.74335001	39.64693846	0.002539689
19	-74.71001662	39.64693846	0.004277129
20	-74.70793328	39.64693846	0.006915997
.....			
95	-74.68501658	39.569855	0.964059916
96	-74.68293324	39.569855	0.967865779
97	-74.7829334	39.56777167	0.971463989
98	-74.78085007	39.56777167	0.97425497
99	-74.68709992	39.56777167	0.977514317
100	-74.68501658	39.56777167	0.980715335

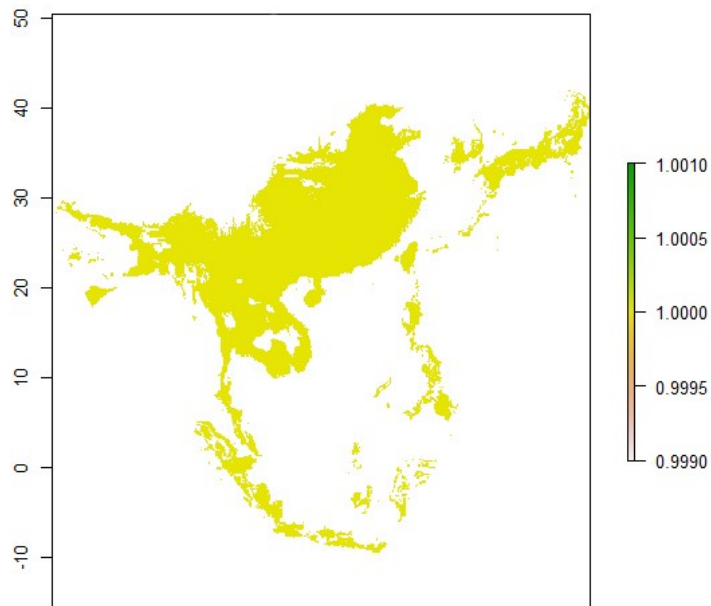
← MTP

← 5th training

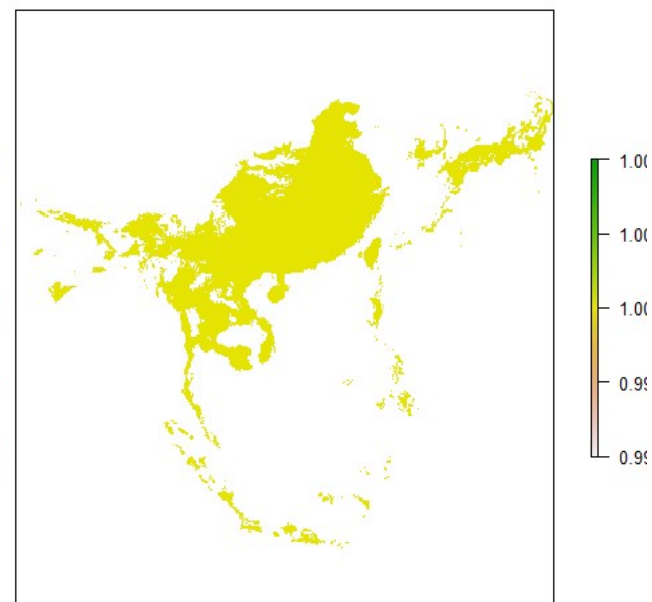
← 10th training

↓

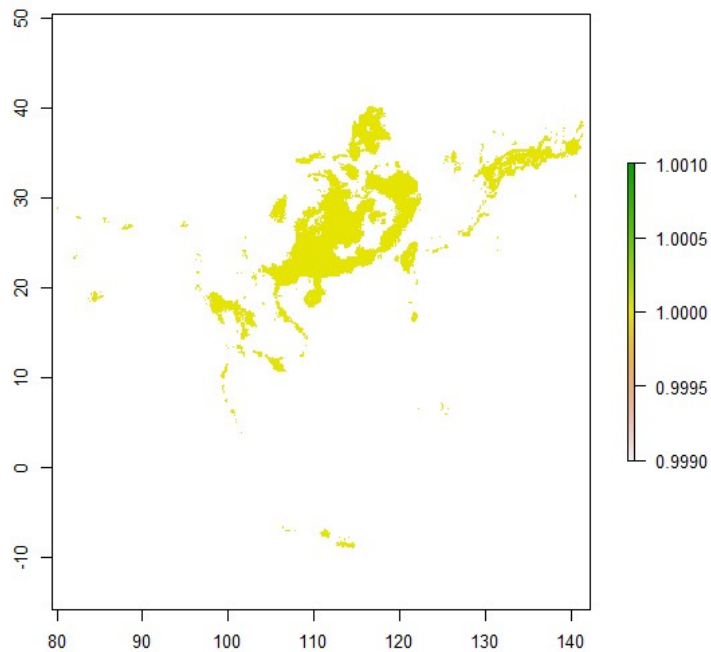
Threshold = 0.1



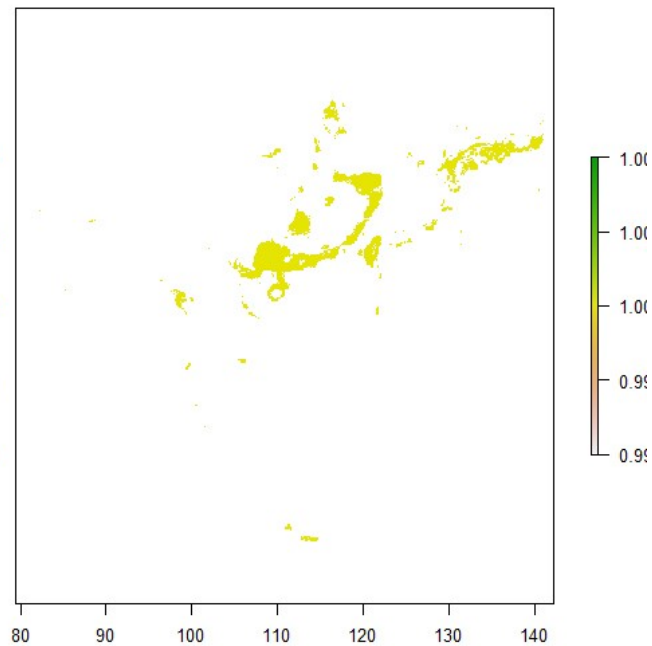
Threshold = 0.2



Threshold = 0.5

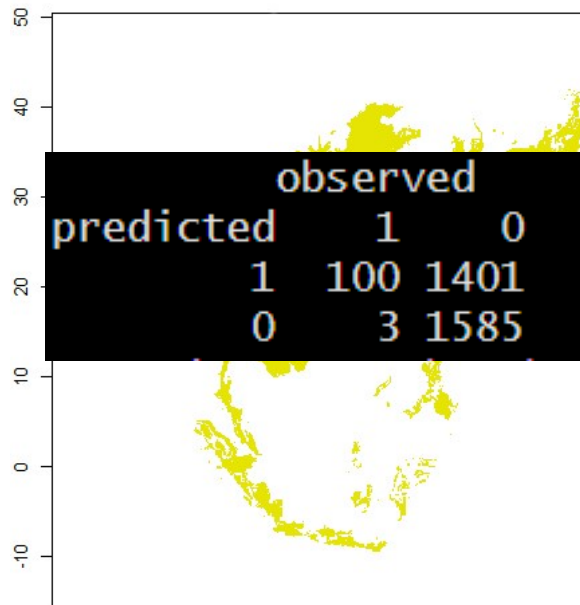


Threshold = 0.7

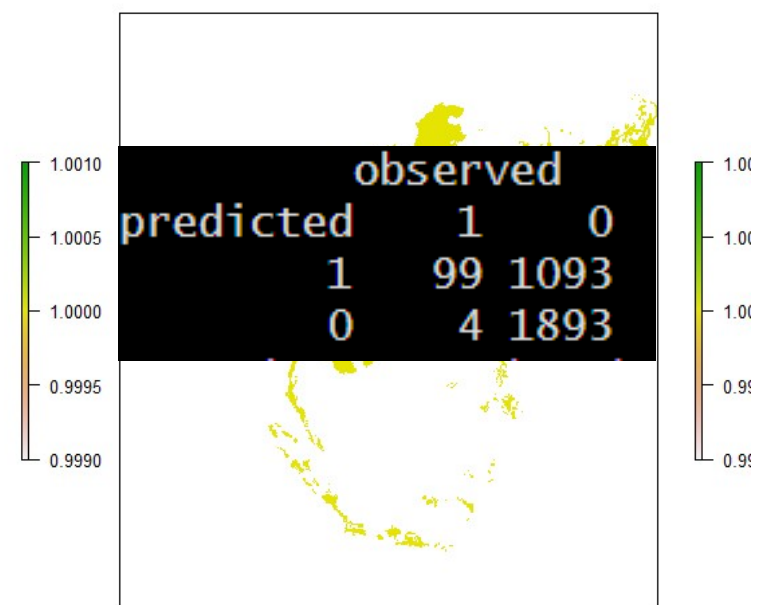


Binary predictions

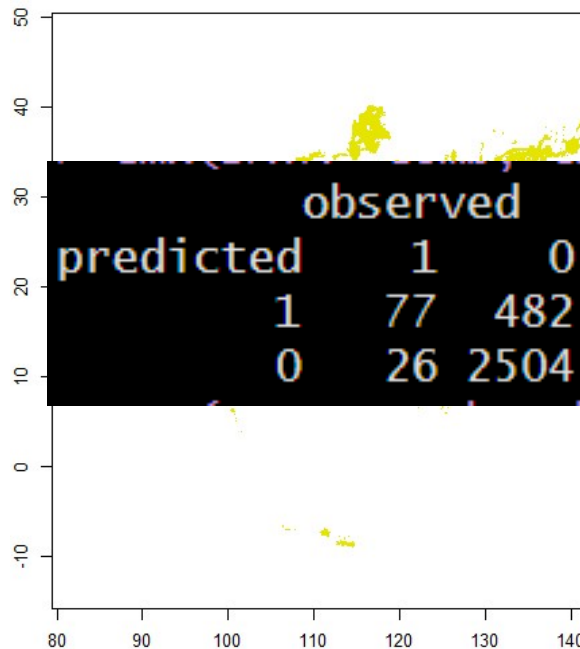
Threshold = 0.1



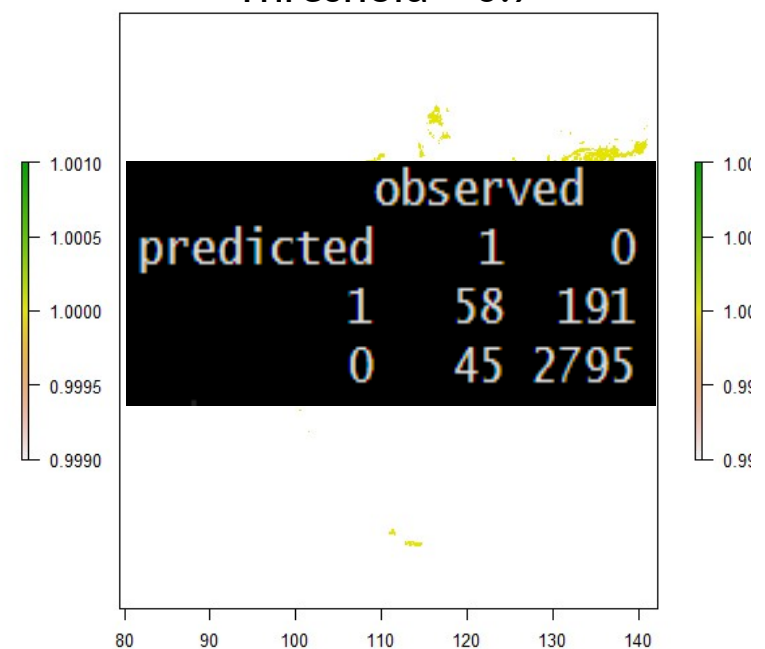
Threshold = 0.2



Threshold = 0.5

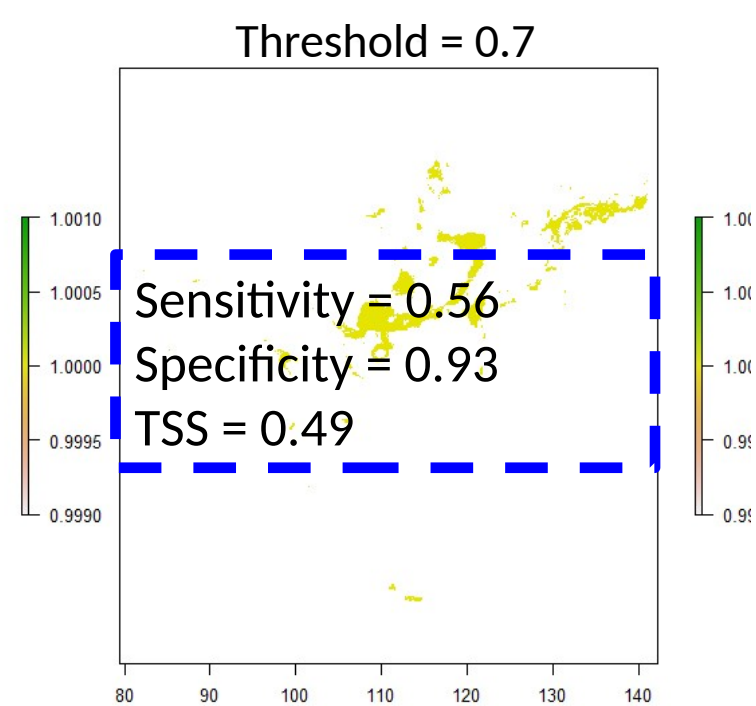
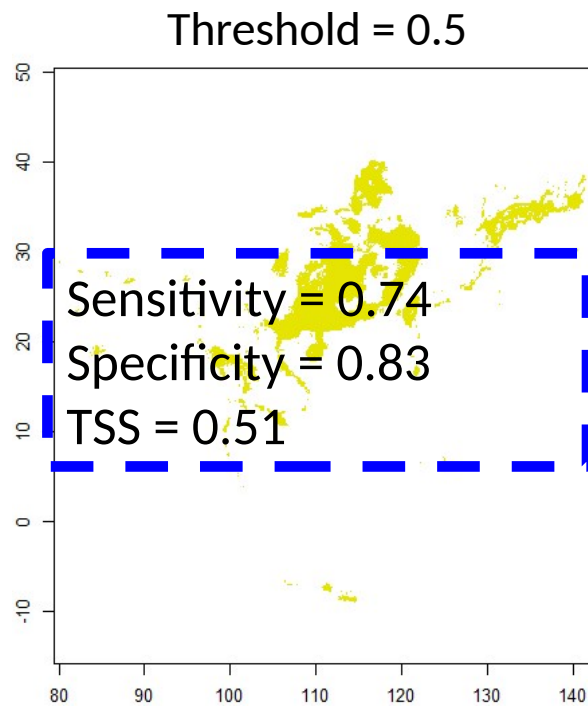
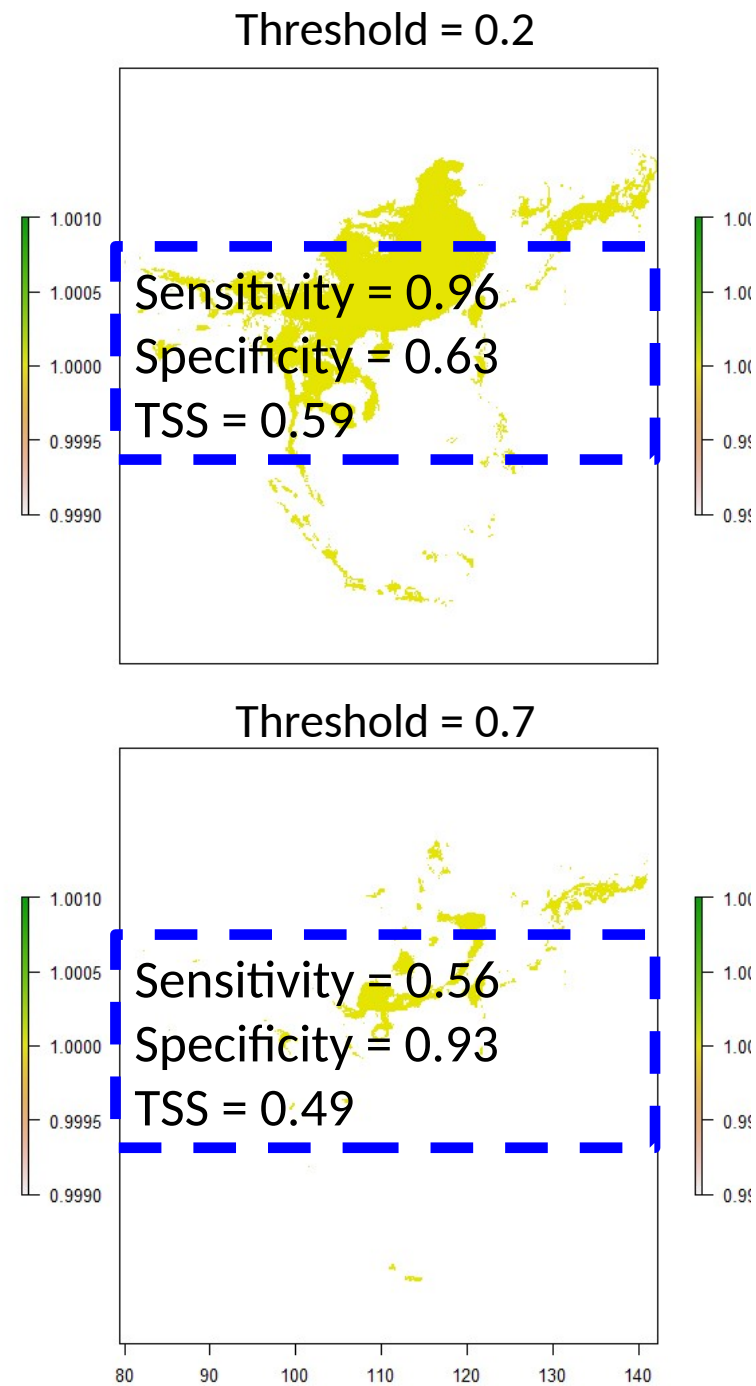
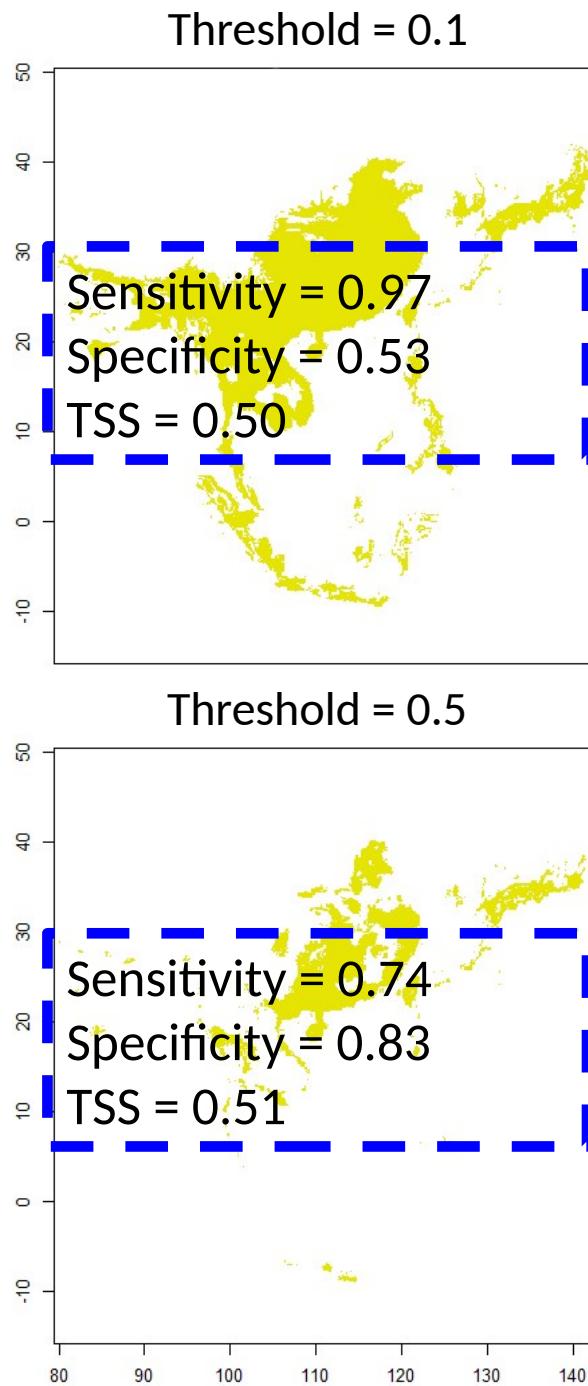


Threshold = 0.7

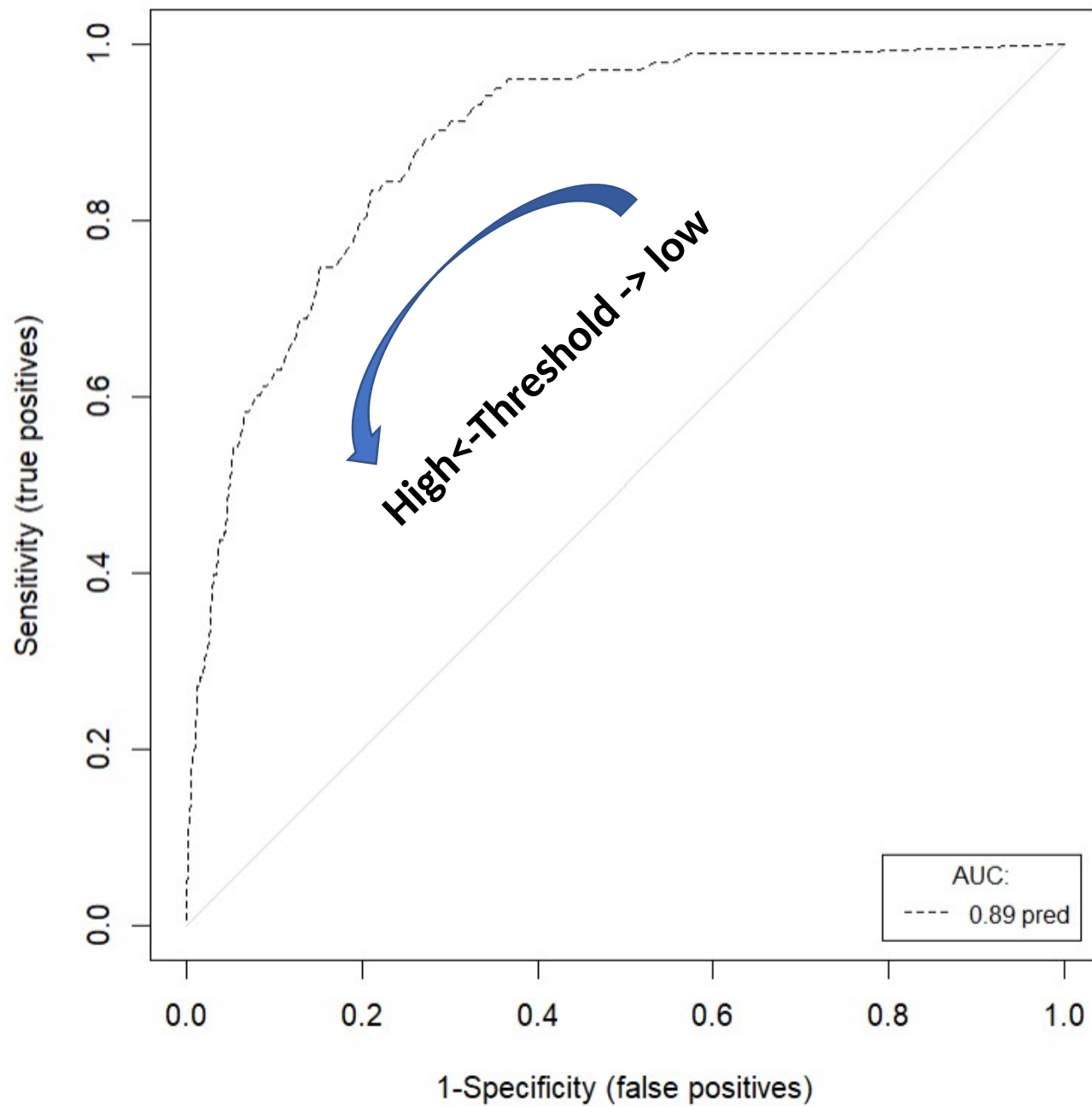


Binary predictions

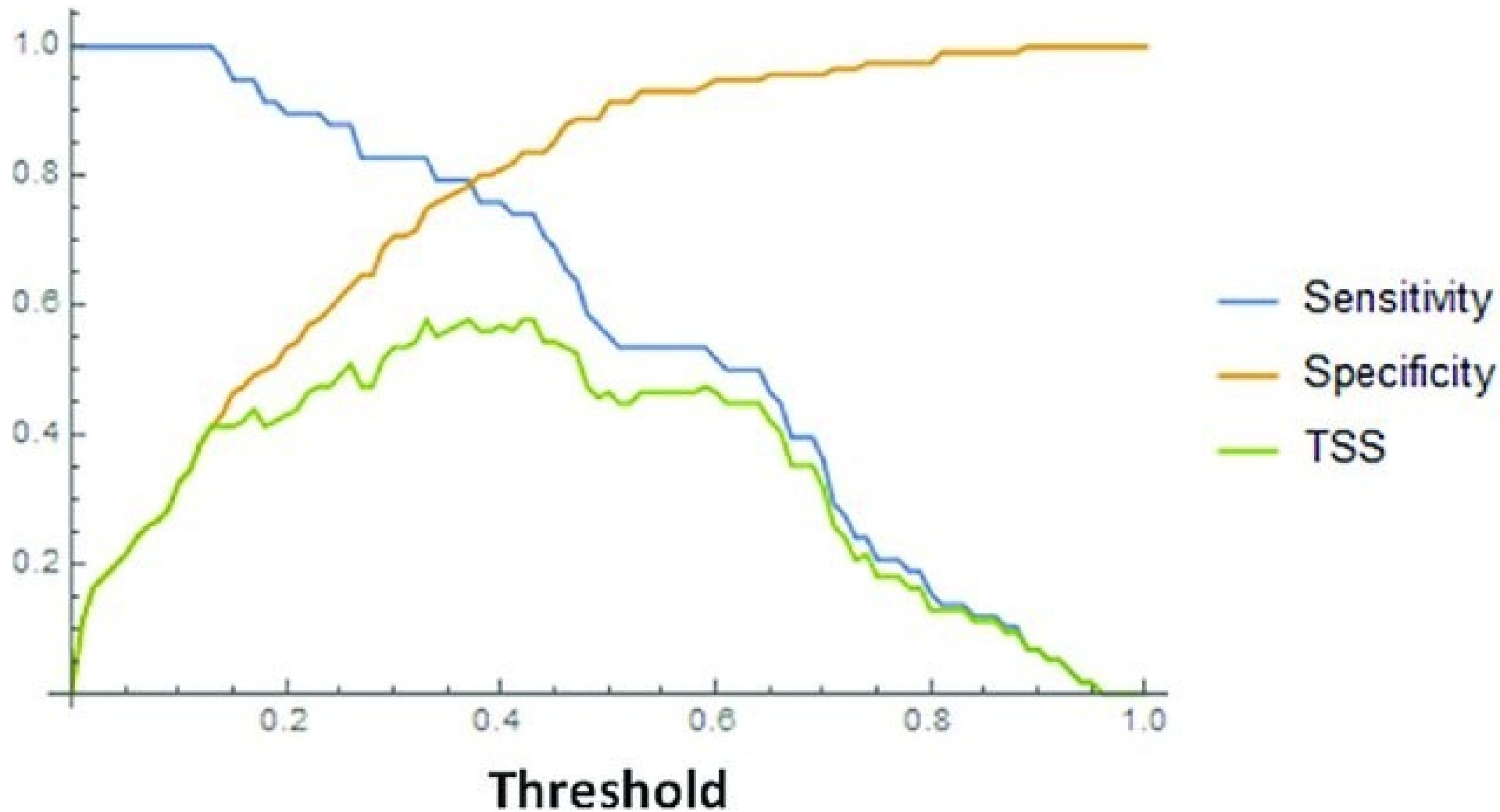
Binary predictions

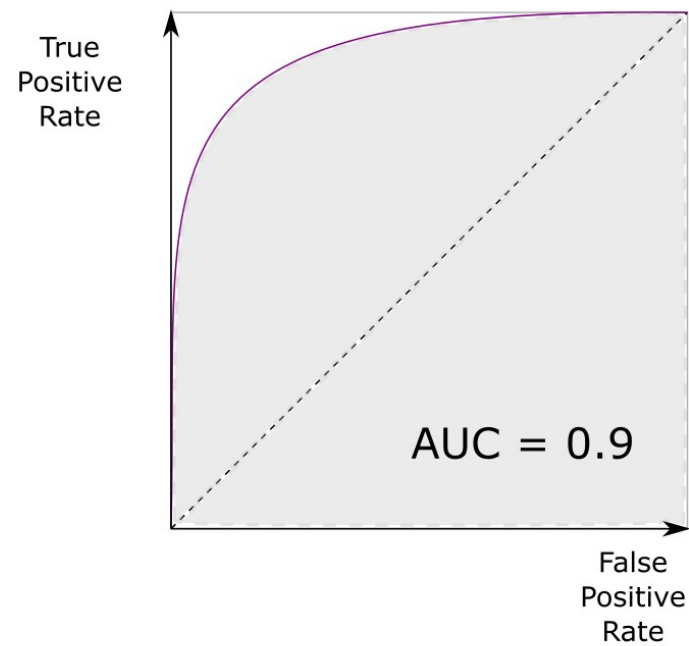
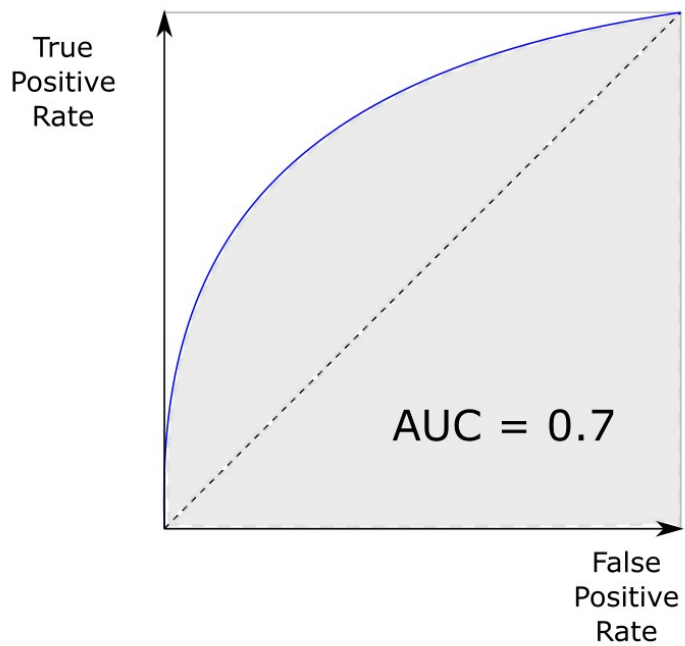
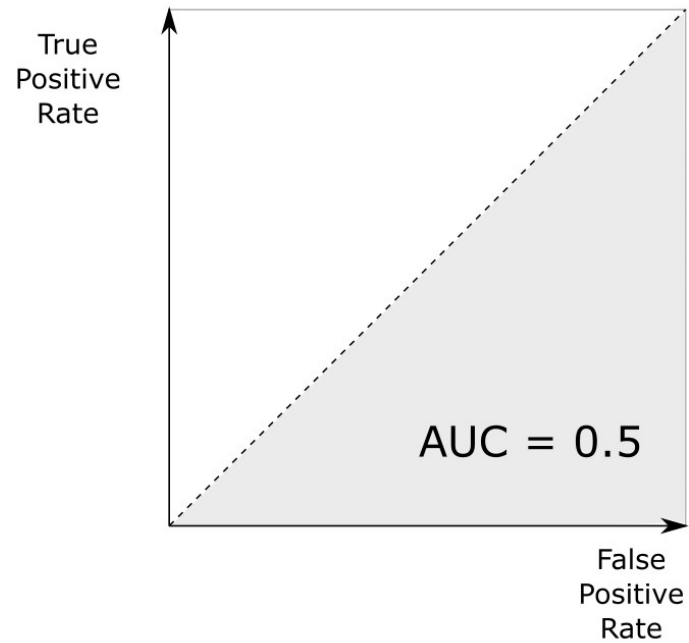
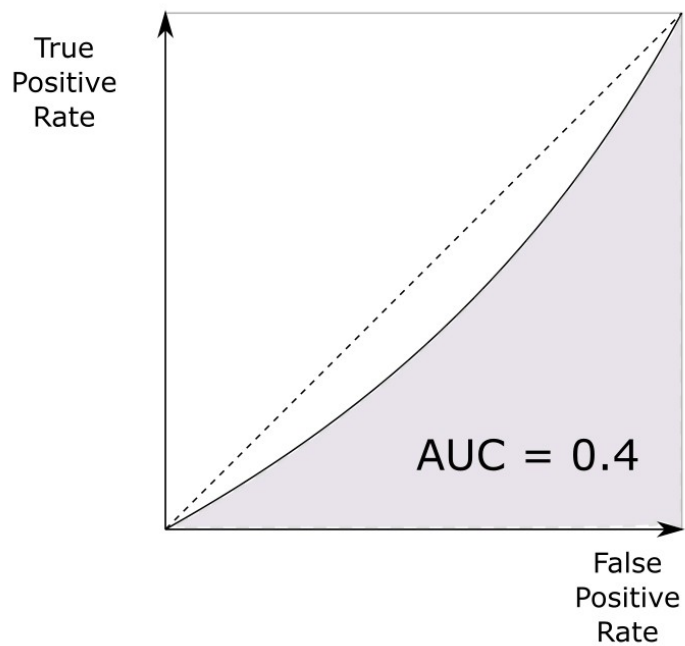


ROC Plot



Sensitivity, specificity, and TSS plotted against threshold values.





Kappa was questioned and TSS was proposed, mainly because of below paper:

*Journal of Applied
Ecology* 2006
43, 1223–1232

METHODOLOGICAL INSIGHTS

Assessing the accuracy of species distribution models: prevalence, kappa and the true skill statistic (TSS)

OMRI ALLOUCHE, ASAF TSOAR and RONEN KADMON

Department of Evolution, Systematics and Ecology, Institute of Life Sciences, The Hebrew University, Givat-Ram, Jerusalem 91904, Israel

Summary

1. In recent years the use of species distribution models by ecologists and conservation managers has increased considerably, along with an awareness of the need to provide accuracy assessment for predictions of such models. The kappa statistic is the most widely used measure for the performance of models generating presence–absence predictions, but several studies have criticized it for being inherently dependent on prevalence, and argued that this dependency introduces statistical artefacts to estimates of predictive accuracy. This criticism has been supported recently by computer simulations showing that kappa responds to the prevalence of the modelled species in a unimodal fashion.



Without quality presence–absence data, discrimination metrics such as TSS can be misleading measures of model performance

Abstract

The discriminating capacity (i.e. ability to correctly classify presences and absences) of species distribution models (SDMs) is commonly evaluated with metrics such as the area under the receiving operating characteristic curve (AUC), the Kappa statistic and the true skill statistic (TSS). AUC and Kappa have been repeatedly criticized, but TSS has fared relatively well since its introduction, mainly because it has been considered as independent of prevalence. In addition, discrimination metrics have been contested because they should be calculated on presence–absence data, but are often used on presence-only or presence-background data. Here, we investigate TSS and an alternative set of metrics—similarity indices, also known as *F*-measures. We first show that even in ideal conditions (i.e. perfectly random presence–absence sampling), TSS can be misleading because of its dependence on prevalence, whereas similarity/*F*-measures provide adequate estimations of model discrimination capacity. Second, we

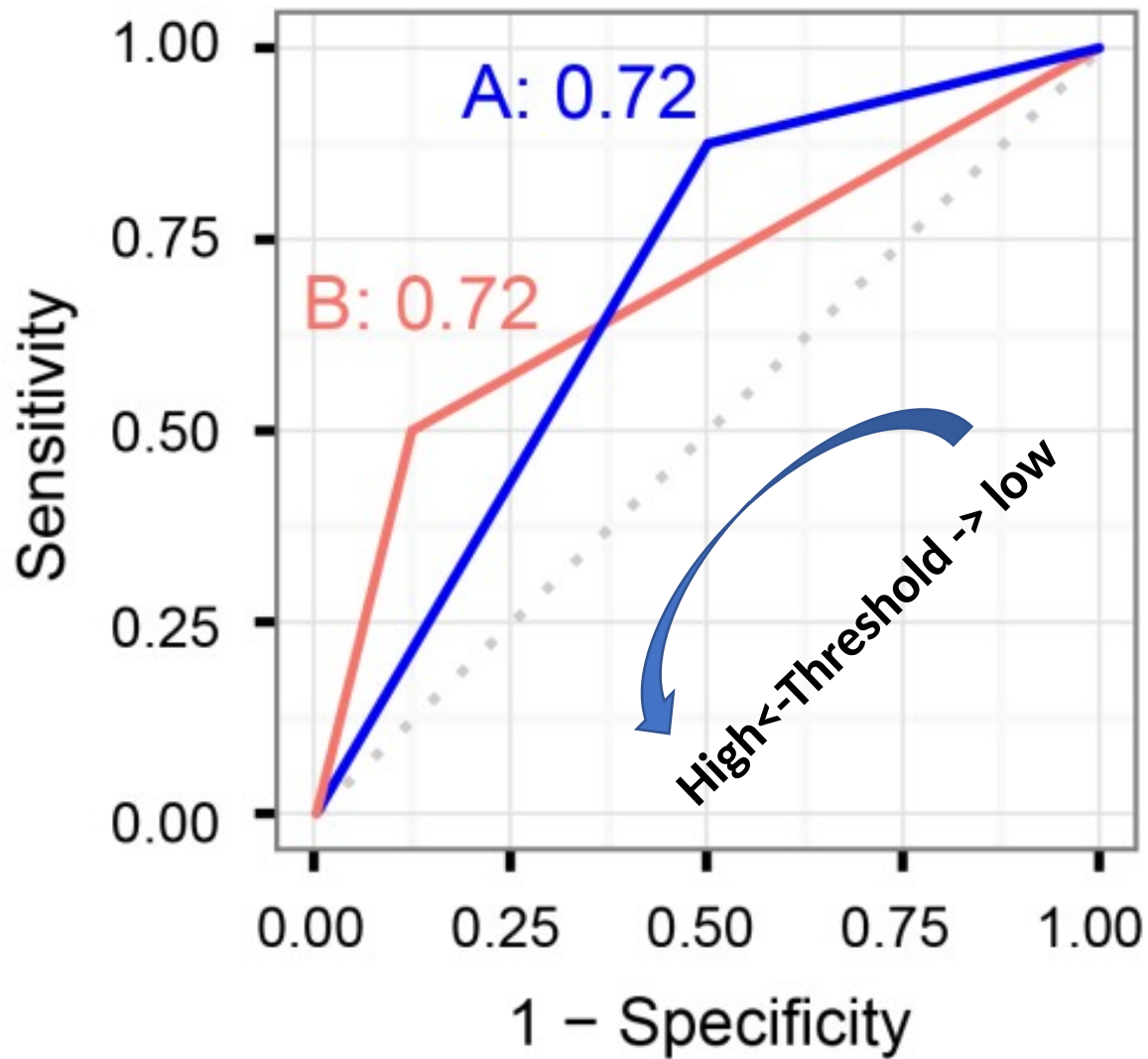
are various methodological choices involved in the application of SDMs (e.g. data type and processing, variables, resolution, algorithms, protocols, global climate models, greenhouse gas emission scenarios), which make them particularly difficult to interpret, compare and assess. However, evaluation of their predictive accuracy is probably a common step to most SDM studies across methodological and technical choices. This evaluation allows us to quantify model performance in terms of how well predictions match observations, which is a fundamental and objective part of any theoretical, applied or methodological study.

To evaluate model predictive performance, the occurrence dataset is often partitioned into two subsets (one for calibrating models, and one for testing) and predictions are assessed in terms of whether or not they fit observations using various accuracy metrics (Araújo, Pearson, Thuiller, & Erhard, 2005), a method called cross-validation. Other approaches include calibrating on the full dataset

Discussion

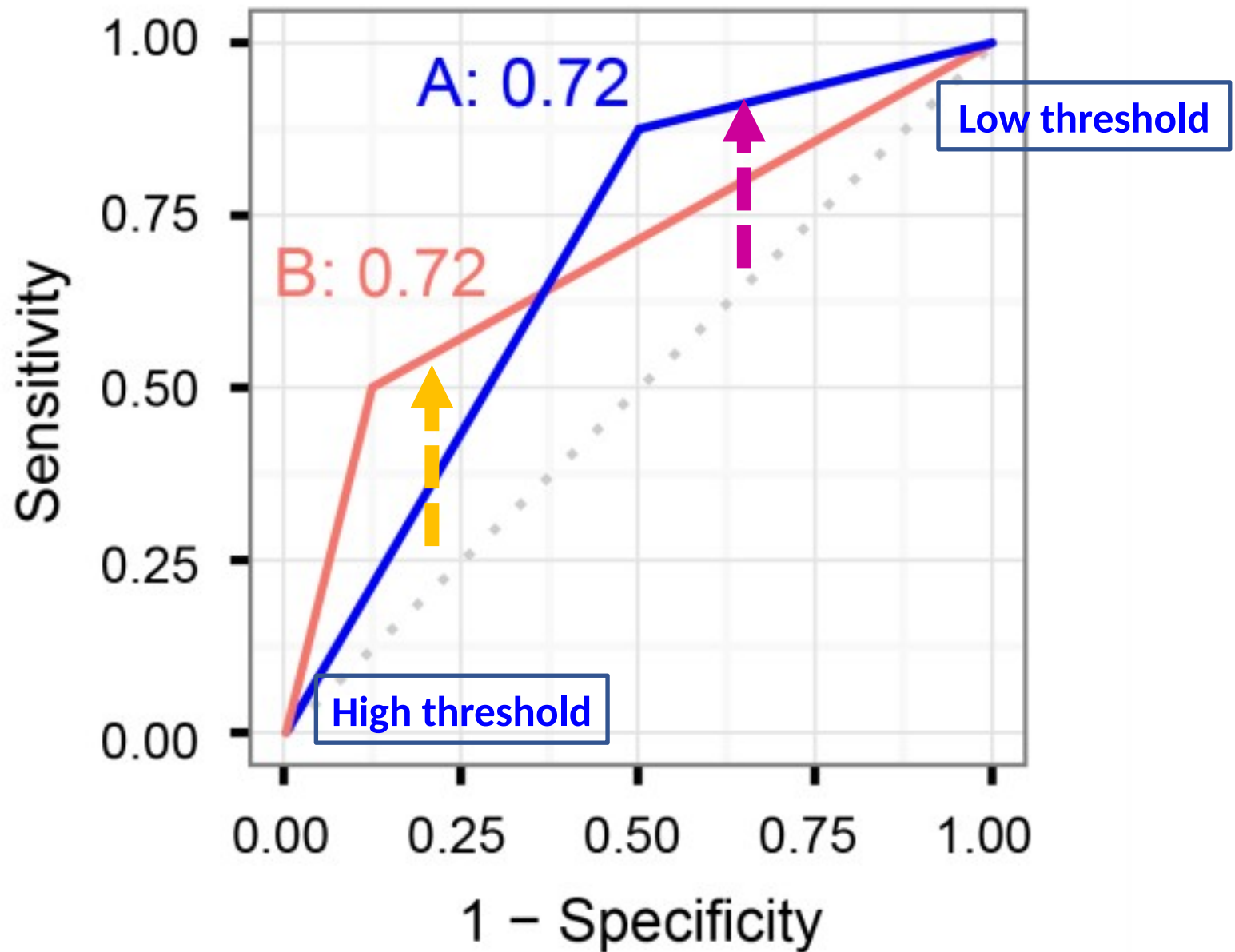
**Are there problems with
AUC ?**

ROC curves with equivalent AUC scores



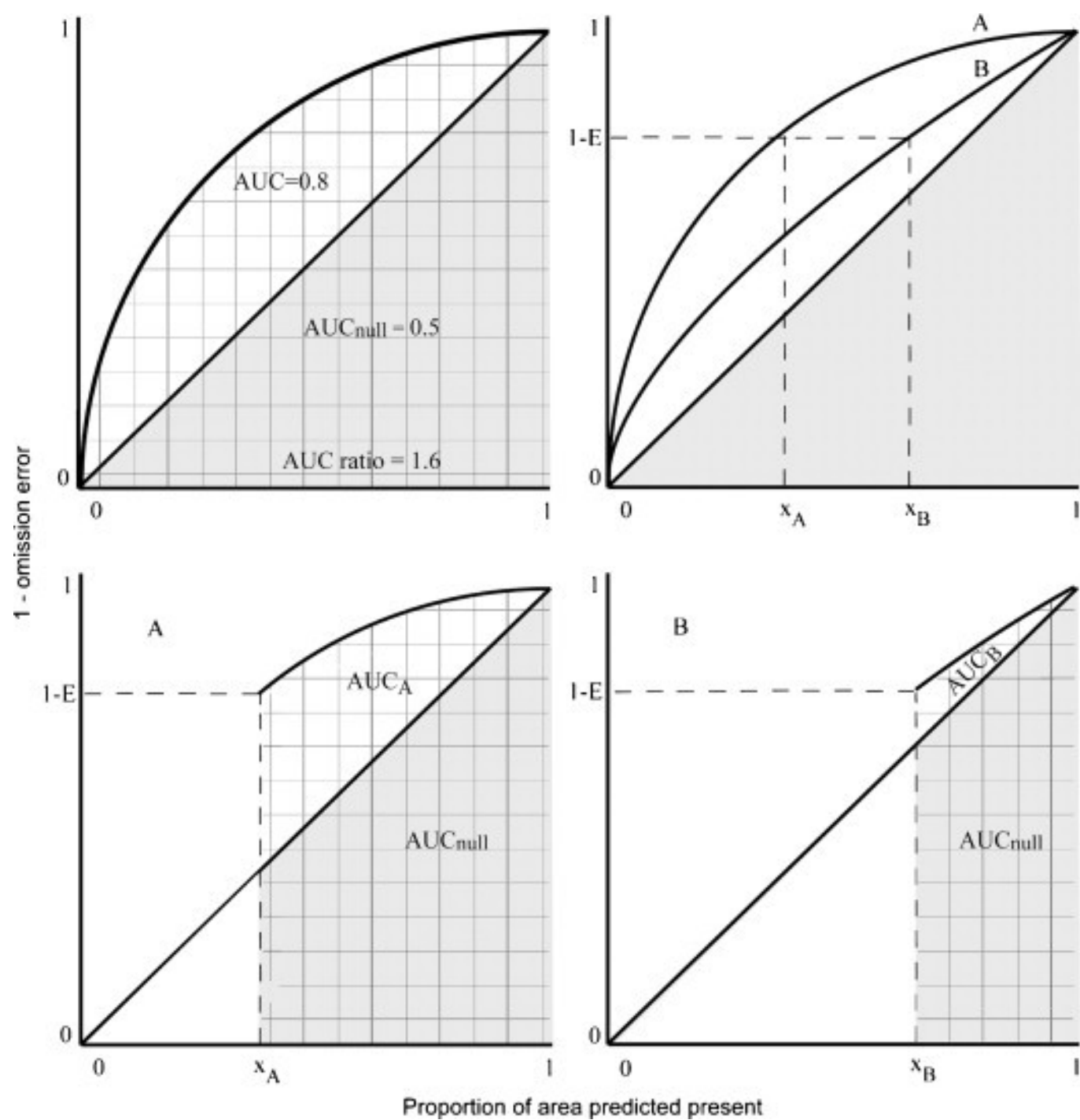
Both “blue” model and “red” model have AUC of 0.72, but are they the same?

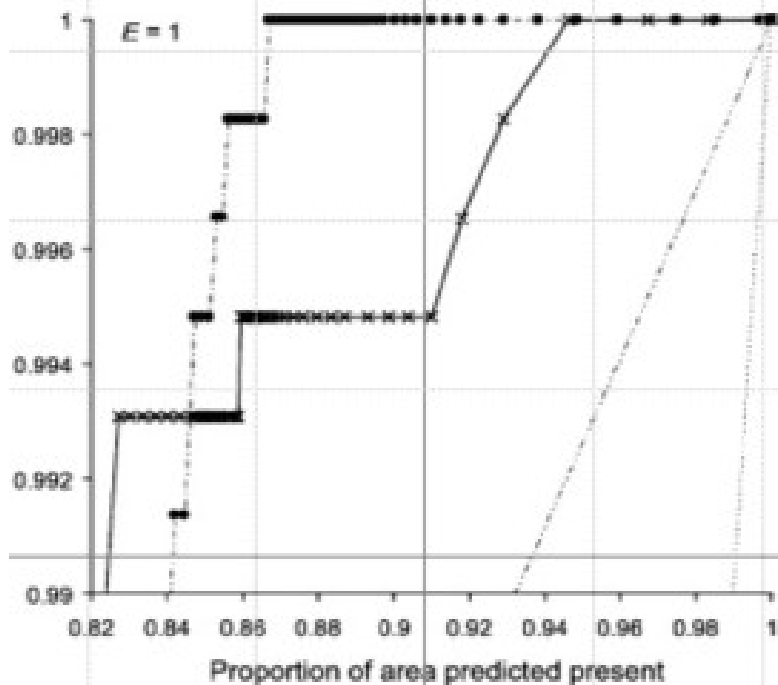
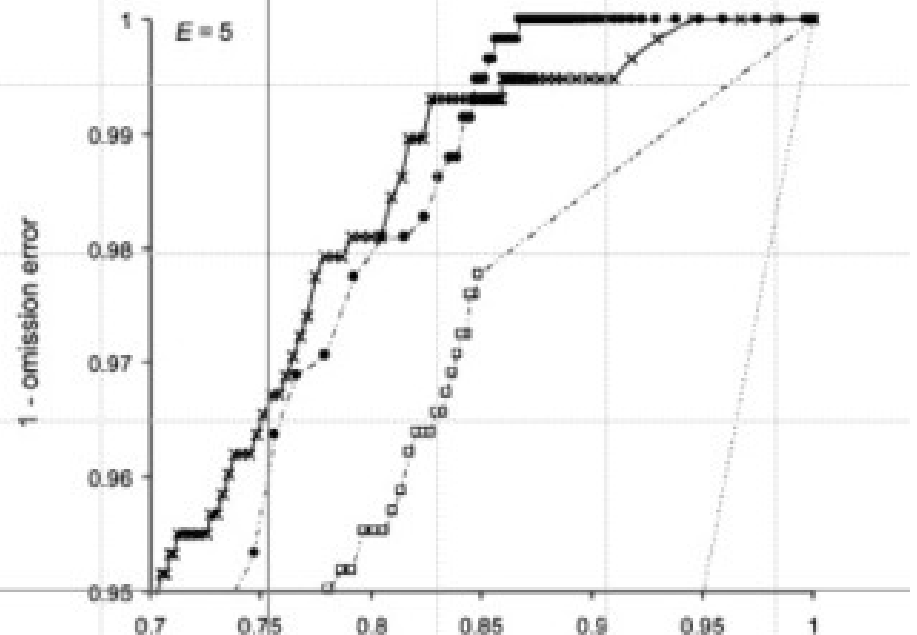
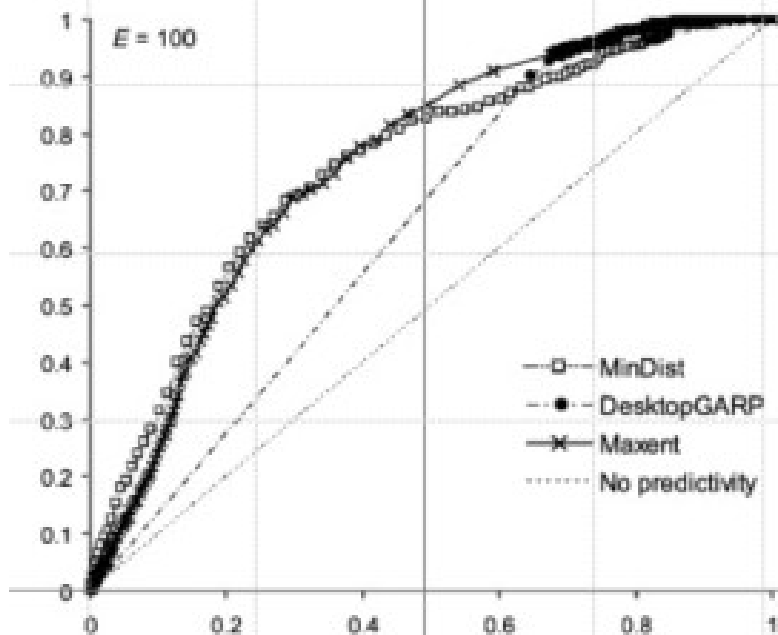
ROC curves with equivalent AUC scores



Partial ROC/AUC ratio

- ◆ ROC has the advantage of being threshold-independent, and as such does not require decisions regarding thresholds;
- ◆ Two errors (omission and commission errors) in AUC should not be weighted equally, as present data are more reliable than absence data;
- ◆ **Partial ROC is a modification of ROC, it use AUC ratio to evaluate model performance**
- ◆ **AUC ratio range 0 to 2, > 1 indicate good model**
- ◆ Partial ROC accounts for occurrence data quality
- ◆ **R package:** ntbox, kuenm



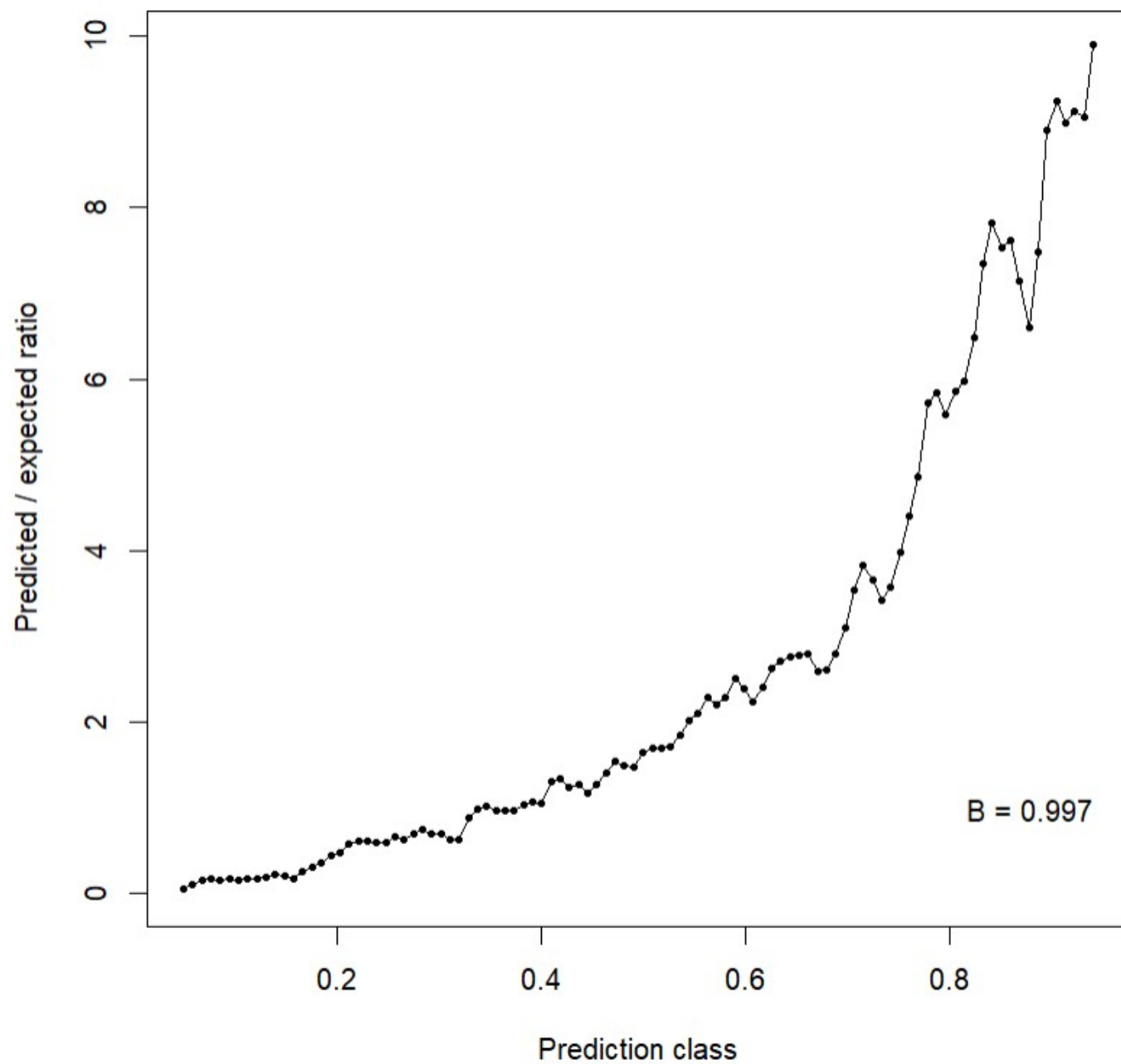


Partial ROC

Boyce index

- ◆ **presence-only metric** for evaluating the predictions of species distribution
- ◆ It measures the predicted-to-expected ratio of presences in each class of predicted values,
- ◆ i.e., whether classes / regions with higher predicted values have indeed higher proportions of presences than expected by chance;
- ◆ **Boyce index ranges: -1 to 1**
- ◆ **Package: ecospat, modEvA**

Boyce index



Demo

kuenm package to calculate Partial ROC-AUC ratio

https://rdr.io/github/manubio13/ku.enm/man/kuenm_proc.html

- `library(kuenm)`
- `data("sp_test", package = "kuenm")`
- `model <- raster::raster(system.file("extdata/sp_model.tif",`
- `package = "kuenm"))`
- `thres <- 5`
- `rand_perc <- 50`
- `iterac <- 500`
- `p_roc <- kuenm_proc(occ.test = sp_test, model = model,`
`threshold = thres, rand.percent = rand_perc, iterations = iterac)`

ntbox package to calculate Partial ROC-AUC ratio

<https://github.com/luismurao/ntbox>

<https://luismurao.github.io/ntbox/reference/pROC.html>

<https://github.com/luismurao/ntbox/blob/master/R/pROC.R>

- # Load a continuous model
- library(ntbox)
- Here I use the same dataset as former kuenm, since I can not install this package

##Original

- partial_roc <- pROC(continuous_mod=conti_model,
- test_data = test_data,
- n_iter=1000,E_percent=5,
- boost_percent=50,
- parallel=FALSE)

###

- partial_roc <- pROC(continuous_mod=model, ##from kuenm
- test_data = sp_data, ##from kuenm
- n_iter=1000,E_percent=5,
- boost_percent=50,
- parallel=FALSE)

ecospat **package to calculate Boyce index**
<https://search.r-project.org/CRAN/refmans/ecospat/html/ecospat.boyce.html>

```
library(ecospat)
```

```
obs <- (ecospat.testData$glm_Saxifraga_oppositifolia  
[which(ecospat.testData$Saxifraga_oppositifolia==1)])
```

```
qq<-ecospat.boyce (fit =  
ecospat.testData$glm_Saxifraga_oppositifolia , obs,  
nclass=0, window.w="default", res=100, PEplot = TRUE)
```

```
qq$cor
```

modEvA to calculate Boyce index

<https://search.r-project.org/CRAN/refmans/modEvA/html/Boyce.html>

<https://www.r-bloggers.com/2022/05/model-evaluation-with-presence-points-and-raster-predictions/>

```
# load sample models:
```

```
library(modEvA)
```

```
data(rotif.mods)
```

```
# choose a particular model to play with:
```

```
mod <- rotif.mods$models[[1]]
```

```
# compute the Boyce index:
```

```
Boyce(model = mod, main = "My model Boyce plot")
```

```
Boyce(model = mod, main = "My model Boyce plot", rm.dup.classes = TRUE)
```

```
# you can also use vectors of observed and predicted values
```

```
# instead of a model object:
```

```
presabs <- mod$y
```

```
prediction <- mod$fitted.values
```

```
Boyce(obs = presabs, pred = prediction)
```

```
# 'obs' can also be a table of presence point coordinates
```

```
# and 'pred' a SpatRaster of predicted values
```